

## CHAPTER 5

# Forces in Nature

Three invisible forces shape our whole world: gravity, electricity, and magnetism.

 Easy Read Edition

### WHAT WE WILL LEARN

- ★ Forces without touching
- ★ Gravity
- ★ Mass vs. weight
- ★ Electric charges
- ★ Conductors and insulators
- ★ Three ways to charge
- ★ Static electricity
- ★ Magnets
- ★ Electromagnets

## 5.1 Forces Without Touching

Drop your phone. It falls. But nothing pushed it. So what made it fall?

**Gravity!** Gravity is a force that pulls things together. It works **without touching**.

There are three forces in nature that can act without touching. We call them **non-contact forces** :



**GRAVITY**  
pulls down



**ELECTRIC**  
between charges



**MAGNETIC**  
magnets and metal

### How Do They Work Without Touching?

These forces work through invisible areas called **fields** .

A field is like an invisible bubble around an object. If you go into that bubble, you feel the force. The closer you are, the stronger the pull.

**Stronger field = stronger force**  
**Closer = stronger. Farther = weaker.**

## 5.2 Gravity

**Gravity** pulls things that have mass toward each other.

Even YOUR mass and the desk's mass pull on each other! But it's so weak you'd never feel it.

Gravity only feels strong when one thing is REALLY big — like Earth.

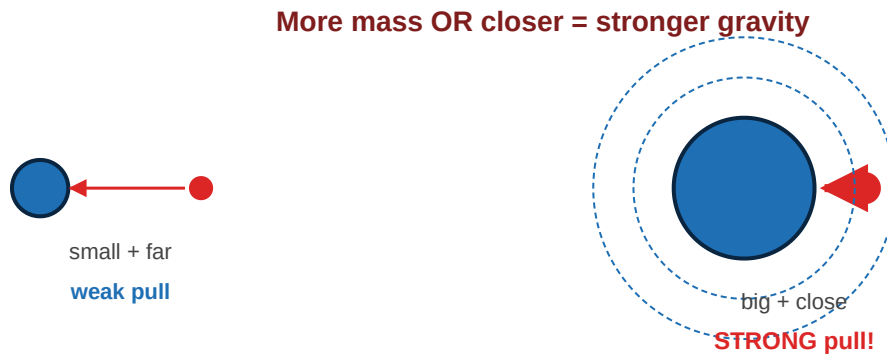
### Two Things Change Gravity

#### 1. Mass

More mass = stronger gravity. Earth has TONS of mass. So it pulls hard.

## 2. Distance

Closer = stronger gravity. The Sun pulls Earth a lot. It pulls Pluto less because Pluto is far away.



*Big + close = strong gravity.*

## 5.3 Mass vs. Weight

People mix these up all the time. They are NOT the same!

### Mass

How much stuff you are made of. Does NOT change. You have the same mass everywhere.

### Weight

How hard gravity pulls on you. DOES change. You weigh less on the Moon!

| Place   | Your Mass | Your Weight |
|---------|-----------|-------------|
| Earth   | 50 kg     | ~110 lbs    |
| Moon    | 50 kg     | ~18 lbs     |
| Mars    | 50 kg     | ~42 lbs     |
| Jupiter | 50 kg     | ~280 lbs!   |

Your mass is the same everywhere. But your weight changes because gravity is different on each planet!

## 5.4 Electric Charges

Remember atoms? They have three parts:

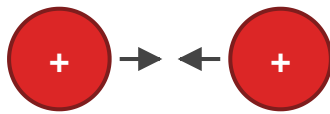
- **Protons** = positive charge (+)
- **Electrons** = negative charge (–)
- **Neutrons** = no charge

Normally, an atom has the same number of (+) and (–). They cancel out. The atom is **neutral**.

But if electrons move around, things become charged!

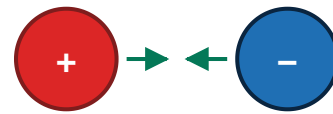
**Like charges PUSH apart.**  
**Opposite charges PULL together.**

**SAME charges → REPEL**



**push apart**

**DIFFERENT charges → ATTRACT**



**pull together**

*Like charges push. Opposite charges pull.*

## 5.5 Conductors and Insulators

Some things let electricity flow through them. Some things block it.

### **Conductors (let it flow)**

Electrons move easily. Like copper, silver, and gold. Most metals!

### **Insulators (block it)**

Electrons can't move freely. Like rubber, plastic, glass, and wood.



**CONDUCTOR**  
copper, silver



**INSULATOR**  
rubber, wood



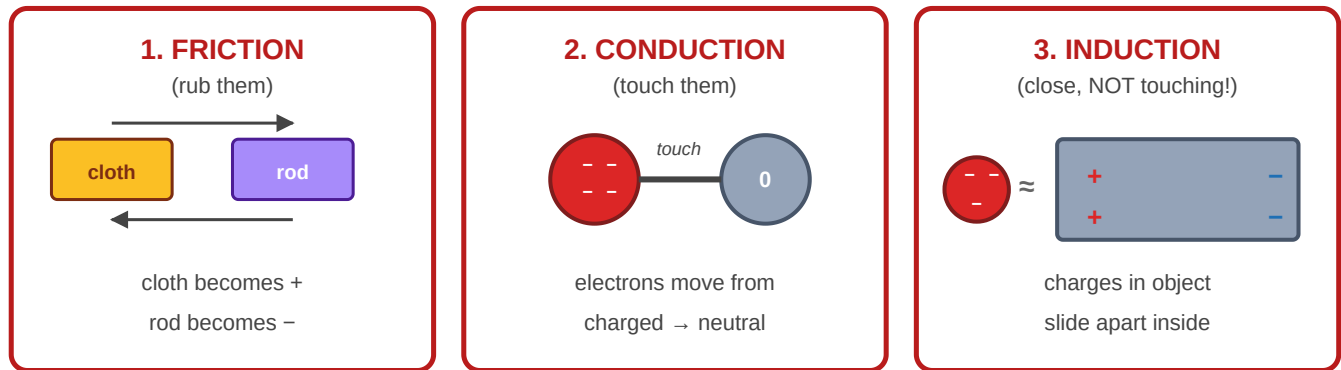
**BOTH**  
in a power cord!

### ★ **Why Wires Have Plastic**

The metal inside a wire is a conductor. Electricity flows through it. The plastic on the outside is an insulator. It stops electricity from jumping into your hand!

## 5.6 Three Ways to Charge Things

How does something become charged? There are three ways. All three move electrons.



*Three ways to charge things up.*

### 1. Friction (Rubbing)

Rub two things together. Electrons jump from one to the other. Like when you rub a balloon on your hair!

### 2. Conduction (Touching)

A charged object touches a neutral one. Electrons flow over. Now both have charge.

### 3. Induction (No Touching!)

A charged object gets close but does NOT touch. It pushes the electrons inside the other object around. No new electrons added — they just move inside.

| Method     | Touch?      | Example                |
|------------|-------------|------------------------|
| Friction   | Yes (rub)   | Balloon on hair        |
| Conduction | Yes (touch) | Shock from doorknob    |
| Induction  | No          | Balloon sticks to wall |

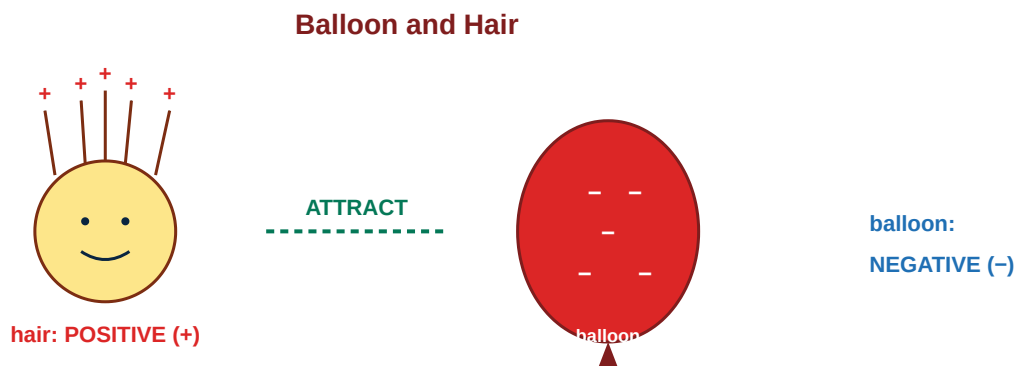
## ★ Your Phone Uses Induction

Your phone screen uses induction! Your finger has a small charge. When you touch the screen, it changes the field below the glass. That's how your phone knows where you tapped!

## 5.7 Static Electricity

**Static electricity** is charge that builds up on things. It happens when you rub things together.

Have you ever walked across a carpet and then touched a doorknob? ZAP! That's static electricity.



*Rub a balloon on your hair. Electrons move from hair to balloon!*

Lightning is just BIG static electricity! Inside a storm cloud, ice crystals rub together. Electrons build up. ZAP — lightning shoots down to the ground.

## 5.8 Magnets

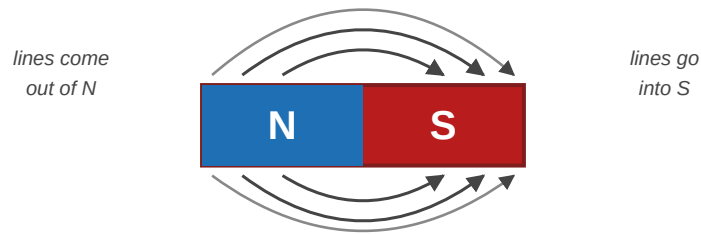
A **magnet** is an object that pulls on iron, nickel, and cobalt.

Every magnet has two ends called **poles**. One is North (N). One is South (S).

**Like poles PUSH apart.  
Opposite poles PULL together.**

Sound familiar? Magnets work just like electric charges!

#### Magnetic Field Around a Magnet



*The magnetic field flows from North to South.*

#### ★ Earth Is a Giant Magnet!

Earth has a hot iron center. It makes a HUGE magnetic field around the whole planet. That's why a compass works — the needle lines up with Earth's field!

This field also blocks dangerous rays from the Sun. Without it, life on Earth couldn't exist!

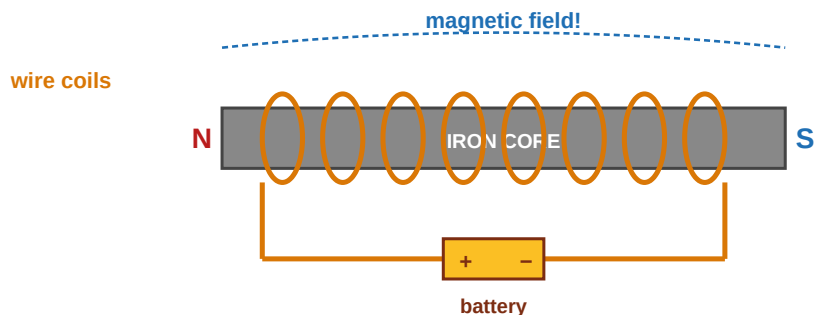
## 5.9 Electromagnets

Here's a wild fact: **moving electricity makes a magnetic field!**

If you wrap a wire around a piece of iron and run electricity through it, you get an **electromagnet** — a magnet you can turn ON and OFF.



## How an Electromagnet Works



*A wire wrapped around iron + battery = magnet you can turn off!*

## How to Make It Stronger

| What to Change    | To Make It Stronger |
|-------------------|---------------------|
| Number of coils   | Wrap MORE wire      |
| Amount of current | Use BIGGER battery  |
| Size of iron core | Use BIGGER iron     |
| Type of core      | Iron works best     |

### ★ Where You See Them

Electromagnets are everywhere!

- Junkyard cranes that pick up cars
- Doorbells and speakers
- Electric motors (fans, blenders)
- MRI machines at the hospital
- Maglev trains that float on tracks!



## The Big Connection

Electricity and magnetism are linked!

- Moving electricity → makes a magnetic field
- Moving magnets → makes electricity flow

This is how power plants make all the electricity for your home. They spin huge magnets near coils of wire. The moving magnets create current!



## Check Yourself

1. Name the three non-contact forces.
2. If your mass is 60 kg on Earth, what is your mass on Mars?
3. Are like poles of a magnet attracted or repelled?
4. Name two ways to make an electromagnet stronger.

## ★ CHAPTER 5 — BIG IDEAS

- Three forces work without touching: **gravity, electric, and magnetic**.
- These forces work through invisible **fields**.
- **Gravity**: more mass = more pull. Closer = more pull.
- **Mass** stays the same everywhere. **Weight** changes with gravity.
- **Like charges repel. Opposite charges attract**. Same for magnetic poles!
- **Conductors** let electricity flow. **Insulators** block it.
- Three ways to charge things: **friction, conduction, induction**.
- Earth is a giant magnet. Its field protects us from the Sun.
- An **electromagnet** is a magnet made with electricity. You can turn it on and off!
- Electricity and magnetism are **linked** — one can make the other.